Statistical tests and assumptions

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Reproducible AI Research Study

THE CONNECTED WEEK

One complete paired-null and exact-tail calculation record.

Keep differences, unit assumptions, null assignments, statistic, ties, p, alpha and interpretation limits together.

Keep one habit: Recall → W1–W3/W5 → independent rescaled case → statistical-test project → delayed return.

MAIN LESSON · SECTION 1

Recall · Retrieve before notes

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

- Without opening the old lesson, explain one key idea from Effect size and practical importance. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Sampling variability. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Baselines and fair comparisons. Give an example and a mistake that the idea helps you avoid.

Effect size and practical importance

My remembered idea, example and correction

Sampling variability

My remembered idea, example and correction

Baselines and fair comparisons

My remembered idea, example and correction

W1 · Explain the current example

> Keep this task’s inputs,rule and response together.

Complete task and evidence

- A statistical test compares an observed statistic with a declared null distribution. Here assume valid independent paired units and conditionally independent equiprobable signs under a symmetric zero-effect null; zero mean alone is not enough. Fix magnitudes and enumerate all sign assignments.

  The main differences of 1, 1, 1 observed differences have absolute mean 1; among eight assignments, all-plus/all-minus have absolute mean 1 and six have 1/3. Inclusive two-sided tail p=2/8=0.25, so do not reject at alpha 0.05. This is not a probability that the null is true or proof of zero effect.

  Separate picture A = [5, 8, 4] and B = [4, 6, 3] seconds gives B–A differences of –1, –2, –1 and absolute sum 4; null signed sums of –4, –2, 0, 2, 4 have counts of 1, 2, 2, 2, 1, again p = 0.25. Exact rational arithmetic retains ties and scale invariance; the earlier absolute-epsilon code is corrected in displayed/default examples. Null enumeration creates no additional observed units.

  Invalid pairing, independence or sign symmetry can invalidate the test despite correct arithmetic. No actual randomization, model study, practical equivalence or causal conclusion is supplied.

- Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

> **My reasoning and evidence**

MAIN LESSON · SECTION 6

W2 · Trace the worked case

> Keep the supplied inputs, comparison rule and evidence together.

Complete task and evidence

- A statistical test compares an observed statistic with a declared null distribution. Here assume valid independent paired units and conditionally independent equiprobable signs under a symmetric zero-effect null; zero mean alone is not enough. Fix magnitudes and enumerate all sign assignments.

  The main differences of 1, 1, 1 observed differences have absolute mean 1; among eight assignments, all-plus/all-minus have absolute mean 1 and six have 1/3. Inclusive two-sided tail p=2/8=0.25, so do not reject at alpha 0.05. This is not a probability that the null is true or proof of zero effect.

  Separate picture A = [5, 8, 4] and B = [4, 6, 3] seconds gives B–A differences of –1, –2, –1 and absolute sum 4; null signed sums of –4, –2, 0, 2, 4 have counts of 1, 2, 2, 2, 1, again p = 0.25. Exact rational arithmetic retains ties and scale invariance; the earlier absolute-epsilon code is corrected in displayed/default examples. Null enumeration creates no additional observed units.

  Invalid pairing, independence or sign symmetry can invalidate the test despite correct arithmetic. No actual randomization, model study, practical equivalence or causal conclusion is supplied.

- Three independent paired differences are 1, 1, 1. Under the stated symmetric zero-effect null, all eight independent sign assignments are equally likely. Use the absolute mean, include equally extreme ties, and compare the exact two-sided p-value with alpha 0.05. Show the intermediate steps and give the conclusion with its scope.

Sign assignment for magnitudes 1, 1, 1	Signed sum	Absolute mean	At least observed 1?
– – –	–3	1	Yes

Sign assignment for magnitudes 1, 1, 1	Signed sum	Absolute mean	At least observed 1?
− − +	−1	1/3	No
− + −	−1	1/3	No
− + +	1	1/3	No
+ − −	−1	1/3	No
+ − +	1	1/3	No
+ + −	1	1/3	No
+ + +	3	1	Yes
Inclusive exact tail	Two of eight assignments	p=2/8=0.25	Above alpha 0.05: do not reject

My reasoning and evidence

MAIN LESSON · SECTION 7

W3 · Repair a misconception

> Keep this task's inputs,rule and response together.

Complete task and evidence

- A statistical test compares an observed statistic with a declared null distribution. Here assume valid independent paired units and conditionally independent equiprobable signs under a symmetric zero-effect null; zero mean alone is not enough. Fix magnitudes and enumerate all sign assignments.

  The main differences of 1, 1, 1 observed differences have absolute mean 1; among eight assignments, all-plus/all-minus have absolute mean 1 and six have 1/3. Inclusive two-sided tail p=2/8=0.25, so do not reject at alpha 0.05. This is not a probability that the null is true or proof of zero effect.

  Separate picture A = [5, 8, 4] and B = [4, 6, 3] seconds gives B–A differences of –1, –2, –1 and absolute sum 4; null signed sums of –4, –2, 0, 2, 4 have counts of 1, 2, 2, 2, 1, again p = 0.25. Exact rational arithmetic retains ties and scale invariance; the earlier absolute-epsilon code is corrected in displayed/default examples. Null enumeration creates no additional observed units.

  Invalid pairing, independence or sign symmetry can invalidate the test despite correct arithmetic. No actual randomization, model study, practical equivalence or causal conclusion is supplied.

- A report omits this qualification: The test requires its sign-**exchangeability** and independence assumptions; do not apply it blindly to clustered or asymmetric differences.

  Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

My reasoning and evidence

W5 · Explain the limitation

> Keep this task’s inputs,rule and response together.

Complete task and evidence

- A statistical test compares an observed statistic with a declared null distribution. Here assume valid independent paired units and conditionally independent equiprobable signs under a symmetric zero-effect null; zero mean alone is not enough. Fix magnitudes and enumerate all sign assignments.

  The main differences of 1, 1, 1 observed differences have absolute mean 1; among eight assignments, all-plus/all-minus have absolute mean 1 and six have 1/3. Inclusive two-sided tail p=2/8=0.25, so do not reject at alpha 0.05. This is not a probability that the null is true or proof of zero effect.

  Separate picture A = [5, 8, 4] and B = [4, 6, 3] seconds gives B–A differences of –1, –2, –1 and absolute sum 4; null signed sums of –4, –2, 0, 2, 4 have counts of 1, 2, 2, 2, 1, again p = 0.25. Exact rational arithmetic retains ties and scale invariance; the earlier absolute-epsilon code is corrected in displayed/default examples. Null enumeration creates no additional observed units.

  Invalid pairing, independence or sign symmetry can invalidate the test despite correct arithmetic. No actual randomization, model study, practical equivalence or causal conclusion is supplied.

- For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

  Do not reuse W4 as a new final test after receiving feedback.

My reasoning and evidence

MAIN LESSON · SECTION 4

Try alone · Selected transfer

> Keep this task’s inputs,rule and response together.

Complete task and evidence

- Three independent paired differences are 2, 2, 2. Assume a symmetric zero-effect null under which each sign flips independently with equal probability. Enumerate all eight assignments. The statistic is absolute mean; include ties as equally extreme. Compute the two-sided exact p-value and interpret at alpha 0.05.

- Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Independent case: Apply a new case independently

My independent explanation, trace and limits

MAIN LESSON · SECTION 10

Project · Keep the research evidence

> Preserve the question, protocol, observations, analysis and limits.

Complete task and evidence

- Add a labelled statistical tests and assumptions **evidence** sheet to Reproducible AI Research Study. Show how you can explain the **mechanism** and terminology of statistical tests and assumptions.

  Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

> **My project change, supporting evidence and remaining limit**

MAIN LESSON · SECTION 11

Feedback · Repair and return

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Supported W2 record**

REFERENCE · SECTION 2

Reference · Supported W2 record

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Use W2: Trace the worked case for Statistical tests and assumptions. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

> **The step I completed without help, and the hint I needed**

▼ Extra practice · Individual reasoning check

REFERENCE · SECTION 3

Extra practice · Individual reasoning check

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Use the worked case for Statistical tests and assumptions. Proposed explanation: "The worked result establishes every future case". Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

> **My explanation before partner discussion**

▼ Reference · Earlier W4 response copy

REFERENCE · SECTION 8

Reference · Earlier W4 response copy

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Three independent paired differences are 2, 2, 2. Assume a symmetric zero-effect null under which each sign flips independently with equal probability. Enumerate all eight assignments. The statistic is absolute mean; include ties as equally extreme. Compute the two-sided exact p-value and interpret at alpha 0.05.

My reasoning and evidence

▼ **Extra practice · Pictured case**

REFERENCE · SECTION 12

Extra practice · Pictured case

> Keep this task's inputs,rule and response together.

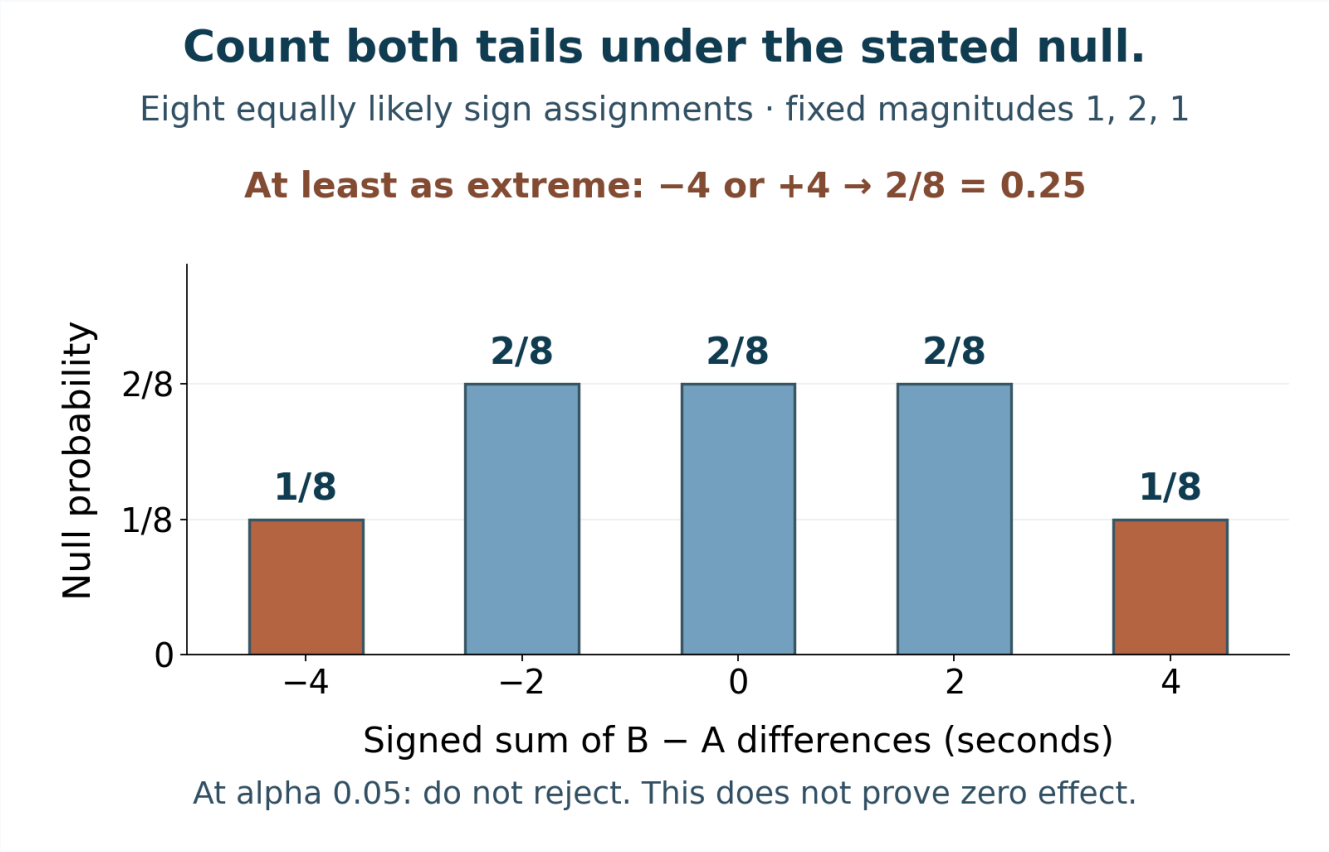
Complete task and evidence

- A small exact paired sign-flip test. Compare three independent task pairs under a null that permits each difference's sign to flip independently with equal probability.

- A times [5,8,4] seconds; B [4,6,3]. Differences B–A are [-1,-2,-1]. Use a two-sided statistic: absolute sum of differences. Assume the stated sign-symmetry null is appropriate.

- Calculate each paired difference and the observed sum. Count null arrangements with absolute sum at least four, compute p, and explain the decision at threshold 0.05.

- Support used: alone / hint / worked example. This record is practice **evidence**.
- Project connection: For the Reproducible AI Research Study, save paired differences, the null assumptions, all eight sign arrangements and the decision rule.



AI ACADEMY / AI-7/14

Three independent paired task times

	A seconds	B seconds	B-A seconds
Task 1	5	4	-1
Task 2	8	6	-2
Task 3	4	3	-1

Read a cell using both its row and column labels.

Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-7/14

Statistical tests and assumptions

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Preserve paired differences

Differences are -1,-2,-1; their sum is -4 and mean is -4/3 seconds.



2. Enumerate null sign flips

Eight equally likely flips give sums -4,-2,0,2,4 with counts 1,2,2,2,1.



3. Count equally extreme sums

Only -4 and 4 have absolute sum at least 4: exact two-sided p=2/8=0.25.

CHANGE ONE THING

Break the task pairing and use unrelated A and B measurements.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

My picture-based explanation and changed-case prediction

▼ Reference · Project milestone

REFERENCE · SECTION 13

Reference · Project milestone

> Keep this task’s inputs,rule and response together.

Complete task and evidence

- Project: Reproducible AI Research Study
- Current milestone: Test changed cases. Keep normal, boundary and failure cases; compare expected and observed results.
- For the Reproducible AI Research Study, save paired differences, the null assumptions, all eight sign arrangements and the decision rule.

Evidence to keep	My record
This week’s input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	
One error or limit and the next test	

Explain what this week adds to your project. Reuse last week's record rather than starting again.

▼ Extra practice · Predict,check and repair

REFERENCE · SECTION 14

Extra practice · Predict,check and repair

> Keep supported feedback separate from the independent case.

First supported question and optional hint

- A small exact paired sign-flip test. Compare three independent task pairs under a null that permits each difference's sign to flip independently with equal probability.
- Which labelled step matches this detail? Only -4 and 4 have absolute sum at least 4: exact two-sided p=2/8=0.25.
- 1. Enumerate null sign flips
- 2. Count equally extreme sums
- 3. Preserve paired differences
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check after the first attempt**

After your attempt, compare with the supported explanation: Count equally extreme sums
Only -4 and 4 have absolute sum at least 4: exact two-sided p=2/8=0.25.

A test compares an observed statistic with outcomes under a specified **null model**. Here the sign-flip assumptions make eight arrangements equally likely.

At a predeclared 0.05 threshold, p=0.25 does not reject the null; it does not prove no difference.

Changed-case question and optional hint

- Changed-case practice: Break the task pairing and use unrelated A and B measurements. Which conclusion is supported by the supplied facts?
- 1. We can decide the result without examining the changed **input** or the procedure.
- 2. This paired sign-flip calculation is no longer justified by the stated design.
- 3. This one example guarantees correct results for every future **input**.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check after the changed attempt**

After your attempt, compare with the supported explanation: This paired sign-flip calculation is no longer justified by the stated design.

Its differences must come from the same task, with the declared independent sign-symmetry assumptions.

Repair and retain the record

Review Effect size and practical importance (ai-7/13). Goal: Explain the **mechanism** and terminology of effect size and practical importance. Short practice: Calculate the absolute error-rate reduction from 20/1000 to 15/1000.

Does it meet the ten-fewer-errors threshold, and what changes when B has eight errors?. Return to this week and explain what you changed.

Review Sampling variability (ai-7/11). Goal: Explain the **mechanism** and terminology of sampling variability. Short practice: Calculate the two four-case rates and their difference.

Explain how changed included cases can affect a fixed model, and why identical deterministic records should not differ on a rerun.. Return to this week and explain what you changed.

Review Baselines and fair comparisons (ai-7/8). Goal: Explain the **mechanism** and terminology of baselines and fair comparisons. Short practice: Calculate 6/8 and 7/8 correctness and name the one-case gain.

Why is giving B the answer key incompatible with an unseen comparison to the simple baseline?. Return to this week and explain what you changed.

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

> **What I repaired, the evidence I kept and the next test**
>
> | | |
> |---|---|
> | | |
> | | |
> | | |
> | | |

▼ **Earlier wording, alternatives and picture transcripts**

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Predict, run and debug

Keep your completed work

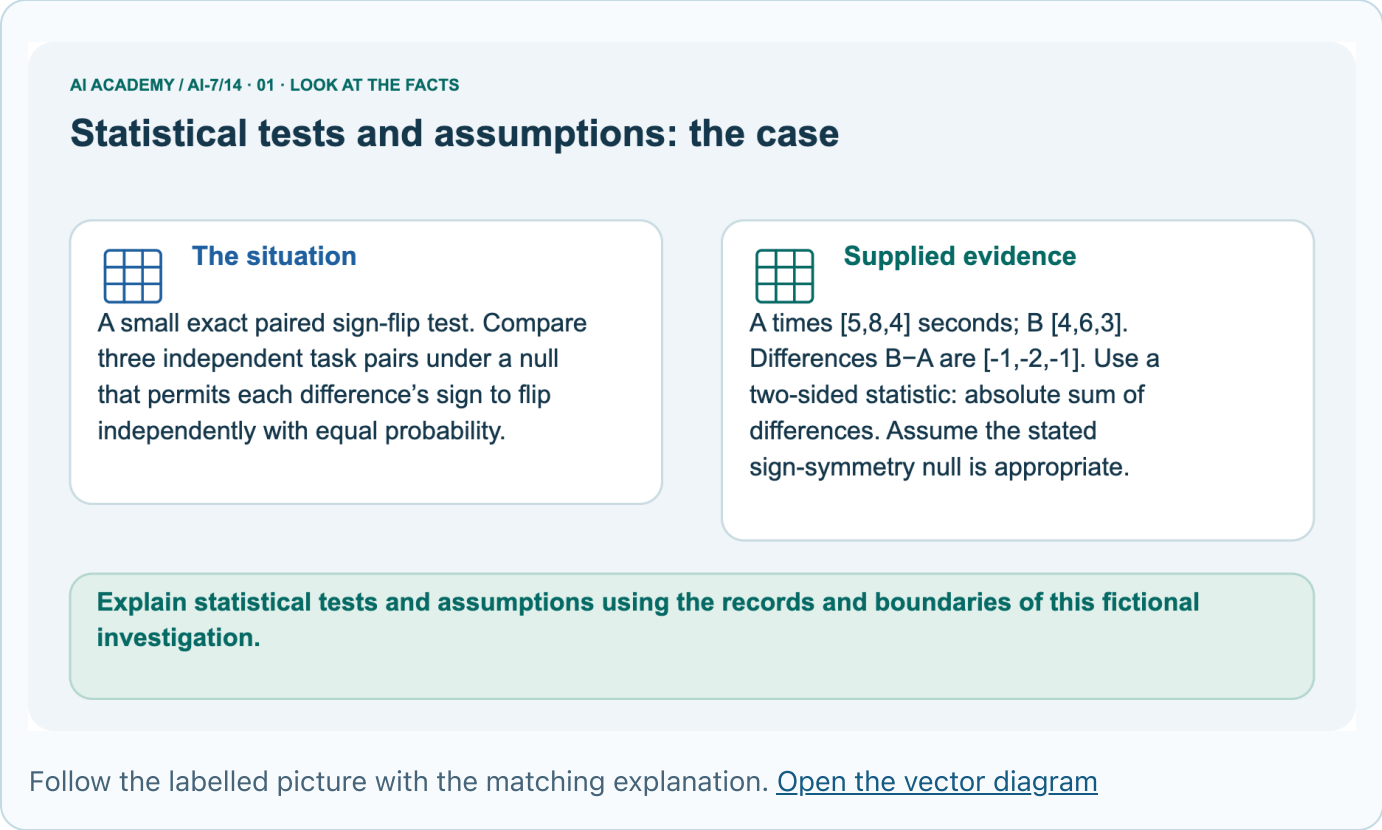
▼ **Reference · Six-picture response sheets**

REFERENCE · SECTION 15

Reference · Six-picture response sheets

> Choose the picture that helps explain the current step.

01. Statistical tests and assumptions: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

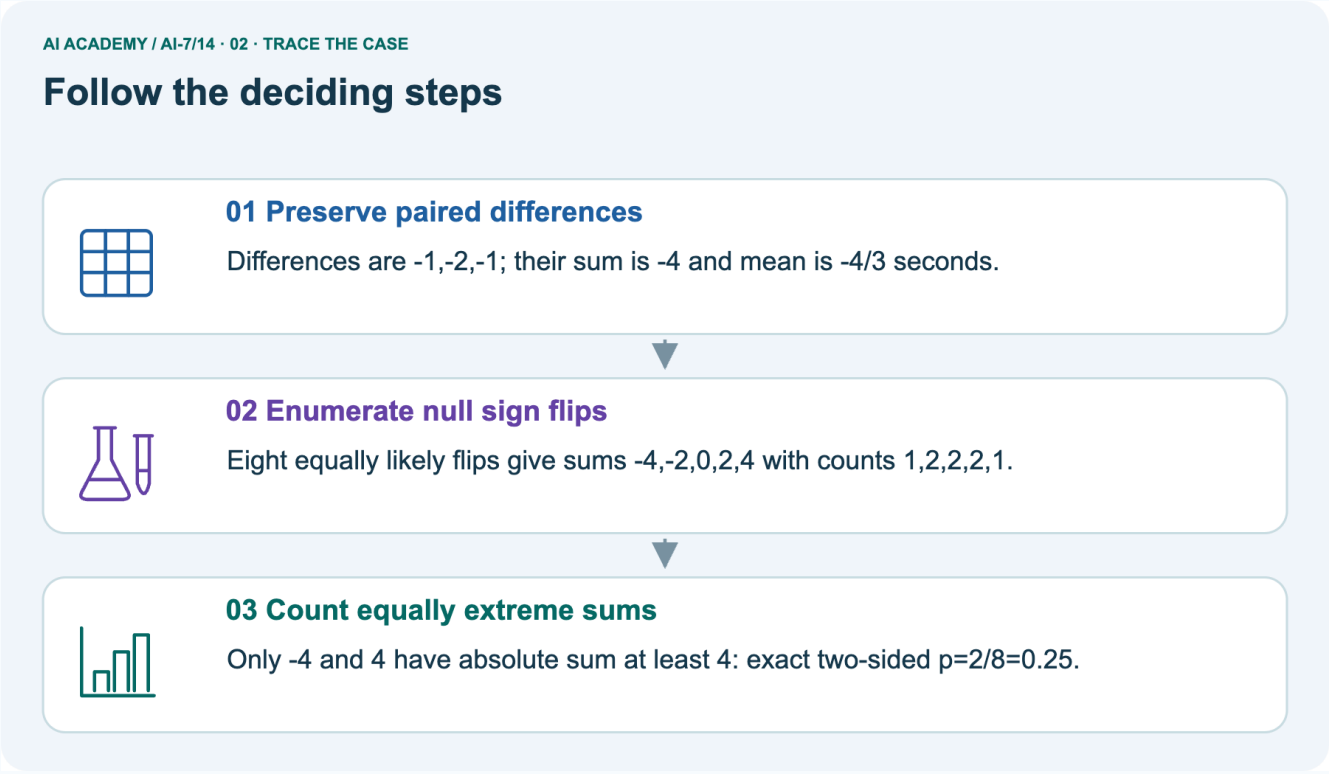
A small exact paired sign-flip test. Compare three independent task pairs under a null that permits each difference's sign to flip independently with equal probability. Supplied facts: A times [5,8,4] seconds; B [4,6,3].

Differences B–A are [-1,-2,-1]. Use a two-sided statistic: absolute sum of differences. Assume the stated sign-symmetry null is appropriate.

Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



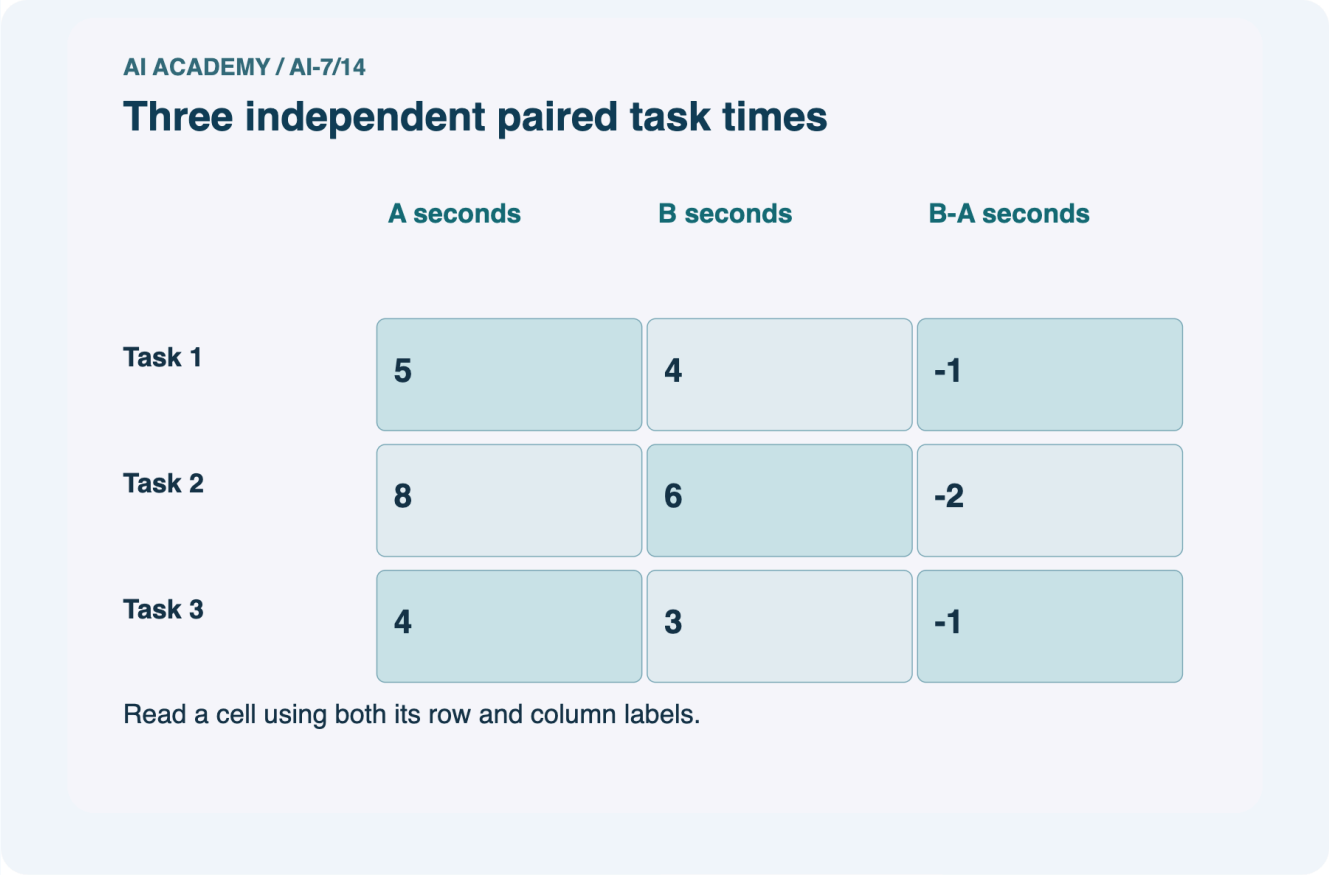
Follow the labelled picture with the matching explanation. Open the vector diagram

Preserve paired differences: Differences are -1,-2,-1; their sum is -4 and mean is -4/3 seconds. → Enumerate null sign flips: Eight equally likely flips give sums -4,-2,0,2,4 with counts 1,2,2,2,1. → Count equally extreme sums: Only -4 and 4 have absolute sum at least 4: exact two-sided p=2/8=0.25.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Three independent paired task times



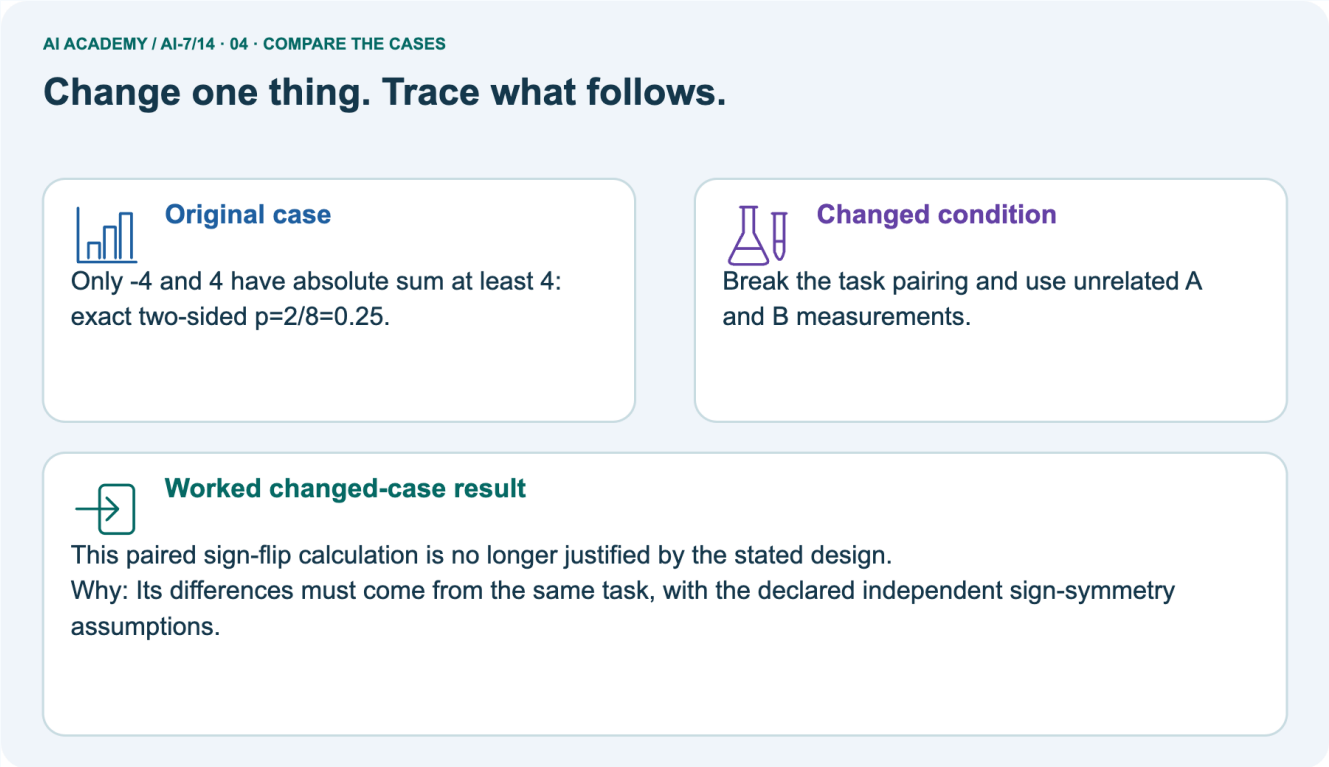
Follow the labelled picture with the matching explanation. Open the vector diagram

Three independent paired task times. Task 1: A seconds = 5, B seconds = 4, B–A seconds = -1. Task 2: A seconds = 8, B seconds = 6, B–A seconds = -2. Task 3: A seconds = 4, B seconds = 3, B–A seconds = -1.

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

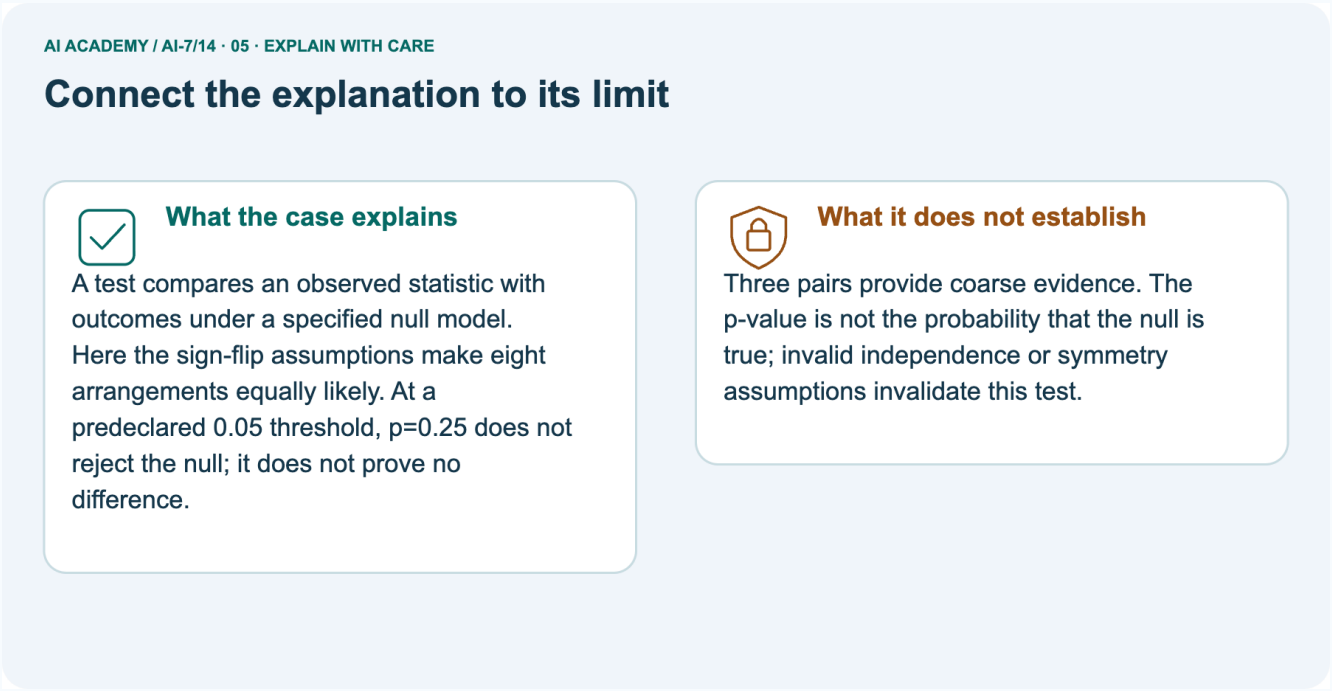
Original case: Only -4 and 4 have absolute sum at least 4: exact two-sided p=2/8=0.25.
Changed condition: Break the task pairing and use unrelated A and B measurements.

Expected result: This paired sign-flip calculation is no longer justified by the stated design.
Reason: Its differences must come from the same task, with the declared independent sign-symmetry assumptions.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

A test compares an observed statistic with outcomes under a specified **null model**. Here the sign-flip assumptions make eight arrangements equally likely.

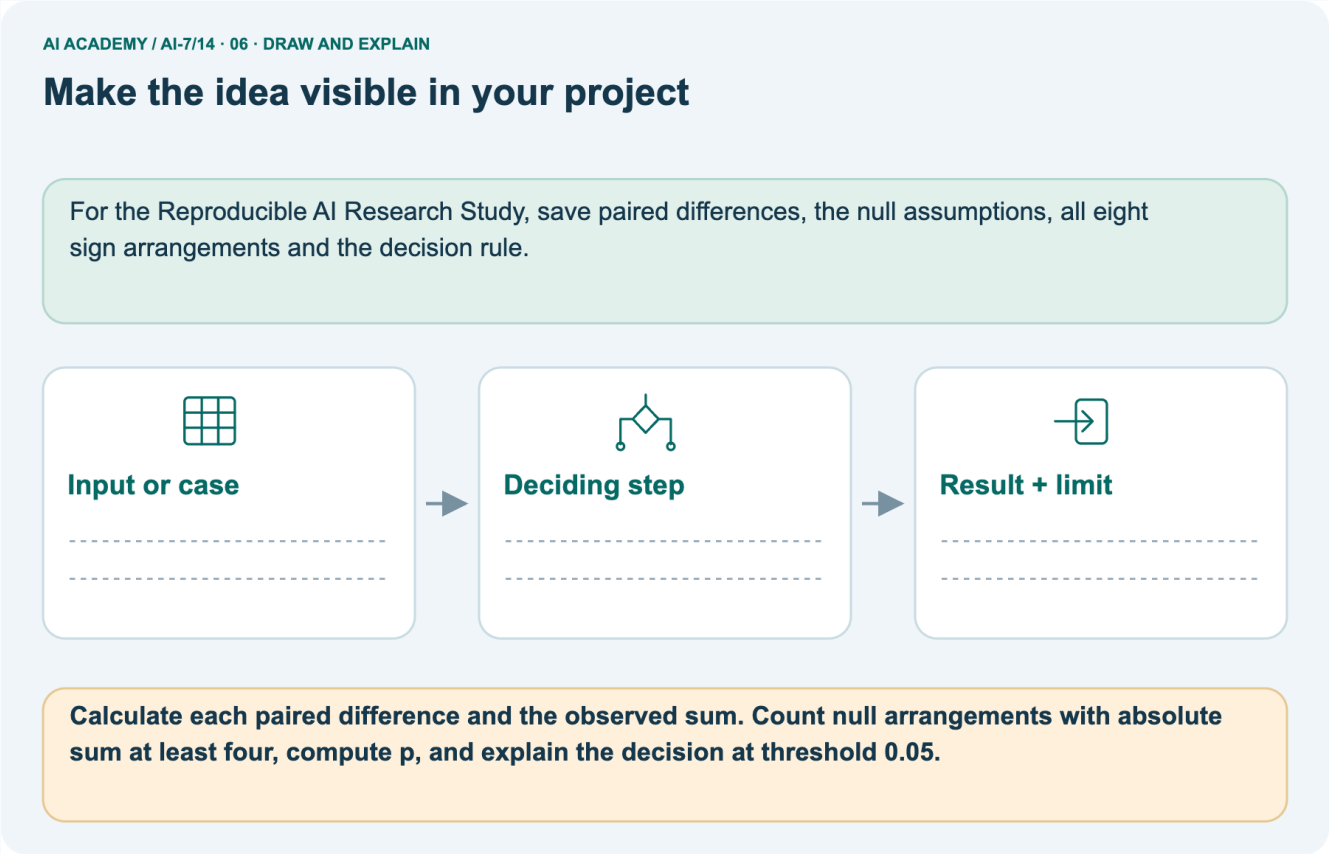
At a predeclared 0.05 threshold, p=0.25 does not reject the null; it does not prove no difference. Limit: Three pairs provide coarse **evidence**.

The **p-value** is not the probability that the null is true; invalid independence or symmetry assumptions invalidate this test.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For the Reproducible AI Research Study, save paired differences, the null assumptions, all eight sign arrangements and the decision rule. Draw your **input**, deciding step, result and limit.

Supported question: Calculate each paired difference and the observed sum. Count null arrangements with absolute sum at least four, compute p, and explain the decision at threshold 0.05.

Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

▼ Earlier wording, alternatives and picture transcripts

01. Statistical tests and assumptions: the case

02. Follow the deciding steps

03. Three independent paired task times

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.